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AI in programmatic advertising: top use cases for publishers in 2025

Explore the applications of AI in programmatic advertising and how to leverage AI for smarter ad monetization for publishers and advertisers.



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Artificial Intelligence (AI) has radically transformed programmatic advertising, driving efficiency and innovation for both the sell side and the buy side. From ad placement optimization to audience targeting, AI's integration into programmatic strategies signifies a leap toward more intelligent, data-driven advertising solutions.

AI frameworks are not just reshaping how ads are bought and sold, it's redefining the very fabric of publisher programmatic and ad monetization, paving the way for more efficient, targeted, and profitable advertising strategies.

The advancement of AI in ad tech is becoming increasingly significant, with diverse applications spanning from buy-side solutions like Trade Desk's [Kokai](#) to the use of generative AI for content creation.

Additionally, AI-based technologies like header bidding and personalized layouts for publishers are emerging, exemplified by solutions that utilize machine learning for traffic shaping and dynamic flooring.

In this article, we'll explore the various facets of AI in programmatic advertising, its main applications, and use cases.

Main applications of AI in programmatic advertising

AI in programmatic advertising plays a pivotal role on both the supply (sell-side) and buy-side, encompassing a wide range of applications. Let's explore some key examples.

Ad stack optimization

Many publishers, SSPs, and DSPs have been using AI to manage and fine-tune their [ad stack](#). AI systems can automate and optimize real-time bidding processes, manage ad inventories, streamline demand partners, and thus maximize ad revenue.

From publishers' perspective, this means having an ad stack that understands user behavior so publishers can decide what bidders include in that auction and increase efficiency, which then increases the advertising spend because the inventory looks of higher quality to the buy side and the

This approach involves dynamically deciding which bidders to include in auctions, thereby improving the overall quality of advertising. The key is to cater to the short attention span of users on specific pages and optimize ad delivery to capture user attention effectively. This strategy not only increases advertising spend but also ensures fulfillment of the advertisers' KPIs.

Personalization and editorial enhancements

Personalization is one of AI's most significant advantages in programmatic advertising. By analyzing user data and delivering more relevant ads on the user level, AI can provide tailored ad experiences to individual users, increasing user lifetime value (LTV), and revenue per session and user, for example.

AI's role in editorial operations also involves analyzing user engagement metrics to guide user experience and content creation, ensuring that editorial output aligns with audience behavior as well as with ad monetization goals. AI tools can suggest topics likely to resonate with readers or optimize content for SEO, thus driving more traffic and engagement.

IA Use Cases for Publishers and Sell-Side

Now, let's delve into the specific use cases for publishers:

Optimizing floor prices for maximum value

AI's dynamic flooring employs sophisticated algorithms and machine learning to decide bottom floor prices and predict the inventory value to inform the buy side. If publishers can transmit more accurate information and influence bids, that can create more efficient trading.

However, fully automated dynamic algorithms are still rare in the industry, offered by only a few publisher tech providers. This involves using AI to accurately predict the value of inventory, thereby informing buyers about the likely bidding price required to secure ad space. The goal is to increase CPM (Cost Per Thousand Impressions) through intelligent pricing strategies.

"This is a relatively advanced method of implementation, which can handle quick changes within traffic CPMs properly so it can adapt quickly, especially to known events but also unknown events", explains Nils Lind, CEO at Assertive Yield.

Traffic shaping

A significant focus is also placed on traffic shaping, where AI identifies high-valued users in real time

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inventory more effectively, reserving high-value spaces for programmatic demand partners and using direct sales campaigns for lower-value programmatic users.

Some [Traffic Shaping](#) Solutions leverage AI to enhance bid rates and filter monetizable requests, effectively managing QPS (Queries Per Second) limits. This optimization strengthens the relationship between publishers, Supply-Side Platforms (SSP), and Demand-Side Platform (DSP) partners.

Additionally, it curtails costs by minimizing invalid traffic. Consequently, these solutions aid in maintaining a specific QPS range with demand partners, ensuring efficient campaign delivery without compromising on overall QPS limits.

Learn more: [How GumGum Achieved. 30% QPS Reduction](#)

Mediating between revenue sources

This involves leveraging publishers' data to decide when to utilize CPA or CPM campaigns or other advertising models. By doing conversions in campaign setup internally, AI can help publishers reduce the margin of error and enhance efficiency, ultimately leading to higher ad revenue.

“I see an opportunity for properly mediating between different revenue sources and types of demand available, not only with Prebid but using the publishers' data to mediate between CPA and CPM campaigns, rather than some external party that has only a fraction of the data available”, Nils Lind explains.

Data integration and unified ad revenue management

Publishers have used AI to integrate data from various sources, creating a unified view of their ad stack. Most ad revenue management tools for publishers gather data from various vendor APIs, a process that can result in discrepancies between GAM (Google Ad Manager) and data from third-party partners.

However, some tools are now utilizing AI and machine learning models to consolidate this data on a single platform, aiming to reduce these discrepancies.

Predictive analytics and revenue predictions

The use of advanced AI and machine learning models not only streamlines data consolidation but also enhances predictive analytics capabilities, enabling publishers to forecast programmatic revenue

In more advanced tools, such as [Yield Manager](#), revenue prediction accuracy is around 94% and 98%.

Optimizing ad placements

AI evaluates user interaction to determine the most effective ad formats and placements, finding a balance between optimal ad revenue and user experience.

It can optimize website layouts to improve ad viewability and CTR without hurting UX by adjusting content and ad placement based on user interaction data, and traffic source, for example.

AI can also understand to some degree how long will the user be there or how they scroll to optimize the ad stack towards getting those impressions and fulfilling the KPIs of the advertiser.

AI content recommendations

In the realm of AI layout personalization, a key feature is dynamic article selection, which utilizes AI to choose the next most relevant article for a user. This process is guided by the user's interests and contextual demand, leveraging Large Language Models (LLMs) that focus on both contextual understanding and potential revenue generation.

From a UX perspective, predictive models can also estimate the user's engagement duration in terms of additional paragraphs read, the probability of the user clicking on content recommendations, and the potential revenue generated per click, user, and session.

This use case leads to numerous benefits, including extended user sessions, increased revenue per session and user, and user lifetime value (LTV).

Use cases of AI in programmatic for advertisers and DSPs

For advertisers and DSPs, AI also offers a multitude of benefits. These include enhanced targeting precision, where AI algorithms identify and segment audiences more accurately.

Also, creative optimization, where AI tests different ad versions to determine which performs best with a particular audience or within particular publishers. Real-time bidding efficiency is another critical area, with AI enabling advertisers to make quicker, more informed bidding decisions based on a comprehensive analysis of available data.

Are you a publisher? Embrace AI in Programmatic

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Assertive Yield is at the forefront of AI for publisher ad revenue optimization. Our expertise in integrating AI frameworks into programmatic strategies offers publishers and SSPs unparalleled benefits.

Assertive Yield's approach to AI in programmatic advertising is not just about squeezing out more revenue; it's about creating transparency and efficiency in the publisher-buyer relationship. Through sophisticated AI tools and strategic planning, publishers can enhance their monetization strategies while fostering more effective advertising ecosystems.

Learn more about the possibilities of unified AI-driven revenue management in our [Next-Gen Publisher Monetization Playbook](#).

Looking for an AI-enabled total ad revenue management tool? [Contact Assertive Yield](#) today to explore how AI can enhance your programmatic strategies and drive your advertising success.

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